

BRAULIO GAEL PORRAS ZUÑIGA

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Data Scientist | Machine Learning Engineer

Professional Profile Data Scientist and Machine Learning Engineer (B.S. in progress, 100% credits completed) with experience in global consulting at Accenture. Specialized in data engineering and modernizing batch architectures into distributed processing systems using Apache Spark. Expert in designing and optimizing data pipelines for analytics and predictive modeling, driving significant operational efficiency. Proven experience in Machine Learning, Deep Learning, and NLP applied to real-world business challenges. Bilingual, Scrum certified, and highly adaptable to new technology stacks.

Technical Skills

Machine Learning & Data Science:

Python (Pandas, NumPy, Scikit-learn), Statistics, Feature Engineering, Model Evaluation.

Deep Learning & NLP:

TensorFlow (Keras), PyTorch, NLP, Computer Vision.

Data Engineering & Big Data:

Apache Spark, ETL Pipelines, Distributed Data Processing, SQL, Java.

Cloud & DevOps:

Google Cloud Platform (GCP), AWS, Docker, Git, Linux, CI/CD.

Methodologies:

Agile Scrum (SDC Certified).

Data Science Projects

Autonomous Financial Compliance Agent (RAG) | Personal Project (AWS) | 2026 | [GitHub](#)

- Developed and deployed a production-grade AI agent (AWS EC2) to automate credit risk analysis and regulatory validation (AML/KYC), reducing manual document review.
- Designed a hybrid RAG architecture using LangChain and GPT-4o, implementing Tool Calling for precise table queries and Guardrails to eliminate hallucinations.
- Built an interactive real-time demo interface using Streamlit.

Tech: Python, LangChain, AWS (EC2), ChromaDB, Streamlit, OpenAI API.

Gold Recovery Optimization Predictive Model (Zyfra) | TripleTen (2025) | [GitHub](#)

- Developed and evaluated a Random Forest regression model to predict gold recovery in industrial settings using a custom sMAPE metric and cross-validation, achieving an sMAPE of 9.36.
- Conducted baseline model comparisons to identify generalization limits and defined improvement opportunities through feature engineering and hyperparameter optimization.

Tech: Python, Scikit-learn, Pandas, Matplotlib.

Customer Churn Prediction | TripleTen (2025) | [GitHub](#)

- Designed and trained classification models to identify high-risk customers, addressing class imbalance through upsampling techniques.
- Optimized F1-Score beyond business baselines, enabling effective segmentation for retention strategies.

Tech: Python, XGBoost, Pandas, Scikit-learn.

Automated Essay Grading System (NLP) | IPN – ESCOM | 2026 (In Development)

- Developing a Deep Learning system for the semantic evaluation of academic texts using NLP techniques and embeddings.
- Functional prototype as part of applied thesis research for degree qualification.

Tech: Python, TensorFlow (Keras), NLP, Deep Learning.

Professional Experience

Accenture, S.C. | Custom Software Engineering Associate | Hybrid | 05/2024 – 05/2025

- Migrated batch data transmission processes to a distributed architecture based on Apache Spark, reducing execution times by approximately 80% through ETL pipeline optimization.
- Implemented CI/CD workflows that improved deployment stability and quality using automated pipelines and DevOps best practices.
- Collaborated in an Agile environment (Scrum), resolving technical debt and contributing to continuous improvement using Jira

Tech: Apache Spark, Python, SQL, Git, CI/CD, Jira.

Education

B.S. in Artificial Intelligence Engineering | IPN – ESCOM | 2020–2026 (Degree in progress).

Data Scientist Professional Certificate | TripleTen | 2025.

Certifications

Scrum Developer Certified (SDC) | SCRUMstudy.

Google Cloud: Build a Secure Google Cloud Network (2025).

Google AI Essentials (2025) & Google Data Analytics Professional Certificate.